

3.2

a) n hledáme ON bázi prostoru $V \rightarrow (u_1, u_2, u_3)$

$$v_1 = 1, v_2 = x, v_3 = x^2$$

$$u_1 = \frac{1}{\|1\|} = \frac{1}{2}$$

$$\hookrightarrow \sqrt{\langle 1, 1 \rangle} = \sqrt{\int_1^5 1 \cdot 1} = \sqrt{4}$$

$$u_2 = \frac{v_2 - w_2}{\|v_2 - w_2\|} = \frac{x-3}{\|x-3\|} = \frac{x-3}{\sqrt{\frac{16}{3}}} = \frac{\sqrt{3}}{4} \cdot (x-3) = 0,433x - 1,299$$

$$v_2 - w_2 = v_2 - \langle u_1, v_2 \rangle \cdot u_1 = x - 6 \cdot \frac{1}{2} = x-3$$

$$= \int_1^5 \frac{1}{2} \cdot x \, dx = 6$$

$$\langle x-3, x-3 \rangle = \int_1^5 (x-3)^2 \, dx = \frac{16}{3}$$

$$u_3 = \frac{v_3 - w_3}{\|v_3 - w_3\|} = \frac{x^2 - 6x + \frac{23}{3}}{\|x^2 - 6x + \frac{23}{3}\|} = \frac{x^2 - 6x + \frac{23}{3}}{\frac{16\sqrt{5}}{15}} = \frac{3\sqrt{5}}{16} \cdot (x^2 - 6x + \frac{23}{3}) = 0,419x^2 - 2,516x + 3,214$$

$$v_3 - w_3 = v_3 - \langle u_1, v_3 \rangle \cdot u_1 - \langle u_2, v_3 \rangle \cdot u_2 = x^2 - \frac{62}{3} \cdot \frac{1}{2} - 8\sqrt{3} \cdot \frac{\sqrt{3}}{4} (x-3) = x^2 - 6x + \frac{23}{3}$$

$$\langle u_1, v_3 \rangle = \int_1^5 \frac{1}{2} \cdot x^2 \, dx = \frac{62}{3}$$

$$\langle u_2, v_3 \rangle = \int_1^5 \frac{\sqrt{3}}{4} (x-3) \cdot x^2 \, dx = 8\sqrt{3}$$

$$\langle x^2 - 6x + \frac{23}{3}, x^2 - 6x + \frac{23}{3} \rangle = \int_1^5 (x^2 - 6x + \frac{23}{3})^2 \, dx = \frac{256}{45}$$

b) Ortogonální projekce $\sin x$ na vektorový prostor s ON bází z minuleho příkladu spočítáme jako $\langle u_1, \sin x \rangle u_1 + \langle u_2, \sin x \rangle u_2 + \langle u_3, \sin x \rangle u_3$

$$\langle \frac{1}{2}, \sin x \rangle = \int_1^5 \frac{1}{2} \sin x \, dx = 0,128$$

$$\langle \frac{\sqrt{3}}{4} (x-3), \sin x \rangle = \int_1^5 \frac{\sqrt{3}}{4} (x-3) \sin x \, dx = -1,483$$

$$\langle \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}), \sin x \rangle = \int_1^5 \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}) \cdot \sin x \, dx = -0,125$$

$$\rightarrow a = -0,053$$

$$b = -0,332$$

$$c = 1,6013$$

$$\text{polynom} = -0,053x^2 - 0,332x + 1,6013$$

c) $\sin x - (ax^2 + bx + c)$ je kolmice na podprostor $\text{Span}\{\vec{u}_1, \vec{u}_2, \vec{u}_3\}$ a jelikož je $(ax^2 + bx + c)$ ort. proj. $\sin x$, tak můžeme využít Pythagorovu větu

$$\|\sin x - (ax^2 + bx + c)\|^2 = \|\sin x\|^2 - \|ax^2 + bx + c\|^2$$

$$\|\sin x\|^2 = \int_1^5 \sin^2 x \, dx = 2,363 \quad \| -0,053x^2 - 0,332x + 1,6013 \|^2 = \int_1^5 (-0,053x^2 - 0,332x + 1,6013)^2 \, dx = 2,2826$$

$$\|\sin x - (ax^2 + bx + c)\| = \sqrt{\|\sin x\|^2 - \|ax^2 + bx + c\|^2} = \frac{\sqrt{2017}}{50} = 0,2835$$